

Unit 11: Metamorphism & Metamorphic Rocks

Reading: pages 151–163 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. Which of the following is *not* one of the five main factors controlling metamorphic processes?
 - a) The color of the parent rock
 - b) The amount of time available for metamorphism
 - c) The mineral composition of the parent rock
 - d) The temperature at which metamorphism takes place
2. Foliation develops in a metamorphic rock as a result of
 - a) the presence of fossils in the parent rock
 - b) confining pressure applied equally in all directions
 - c) directed pressure or shear stress
 - d) very rapid cooling
3. Which list places the foliated metamorphic rocks in order of *increasing* metamorphic grade?
 - a) Phyllite, slate, gneiss, schist
 - b) Schist, gneiss, slate, phyllite
 - c) Gneiss, schist, phyllite, slate
 - d) Slate, phyllite, schist, gneiss
4. Marble is metamorphosed
 - a) limestone
 - b) shale
 - c) basalt
 - d) sandstone
5. Water matters a great deal in metamorphism chiefly because it
 - a) cools the rock and slows the reactions down
 - b) speeds reactions up by helping ions move, and carries dissolved elements from place to place
 - c) prevents foliation from forming
 - d) dissolves all the minerals present

6. In most areas, the normal geothermal gradient — the rate at which temperature rises with depth — is about
- a) 3°C per kilometer
 - b) 30°C per kilometer
 - c) 100°C per kilometer
 - d) 300°C per kilometer

Part B · True or False

2 questions × 1 point = 2 points

7. Quartzite and marble often show no foliation even when they form under directed pressure, because quartz and calcite do not tend to align.

TRUE / FALSE

8. Contact metamorphism typically affects a zone tens of thousands of square kilometers across.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain the difference between *contact* and *regional* metamorphism. Where does each occur, and is the resulting rock usually foliated?

10. Blueschist forms in large quantities, yet it is almost never seen at Earth's surface. Explain why, and name one place where it *can* be seen.

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